

Abhishek Ajith

PhD Researcher in Regenerative Biomaterials

Regenerative Biomaterials and Theranostics (RBT) Lab, Cochin University of Science and Technology (CUSAT)

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PROFILE

I design inherently radiopaque, biodegradable biomaterials — implants that are visible under X-ray, osteogenic, and resorb after healing — for orthopaedic repair and locoregional cancer therapy.

EDUCATION

PhD — Regenerative Biomaterials and Theranostics

Cochin University of Science and Technology · Ernakulam, Kerala, India · 2026 - Ongoing

Doctoral research on radiopaque biodegradable biomaterials for orthopaedic repair and locoregional cancer therapy.

M.Tech. Polymer Technology; CGPA 8.52/10.00

Cochin University of Science and Technology · Ernakulam, Kerala, India · Aug 2020 - Aug 2022

Secured 1st position in the 2020–22 batch.

M.Sc. Biopolymer Science; CGPA 8.76/10.00

CIPET: Institute of Petrochemicals Technology · Ernakulam, Kerala, India · July 2017 - Jun 2019

B.Sc. Chemistry; CGPA 8.70/10.00

Mahatma Gandhi University · Kottayam, Kerala, India · July 2014 - May 2017

EXPERIENCE

Junior Research Fellow

Department of Polymer Science and Rubber Technology, CUSAT · Ernakulam, Kerala, India · July 2023 - March 2026

Worked on "Development of Inherently Radiopaque Biodegradable and Cost Effective TACE System with Liver Regeneration Potential for The Treatment of Hepatocellular Carcinoma" under Prof. G. S. Sailaja. Developed a natural radiopaque component using nanocellulose extracted from Agave sisalana and prepared drug-eluting chemotherapeutic porous microspheres.

Project Officer

Department of Metallurgical and Materials Engineering, IIT Madras · Chennai, Tamil Nadu, India · Nov 2022 - July 2023

NPOL–DRDO funded project. Prepared specialty interpenetrating polymer network (IPN) coatings for underwater sound damping. Characterised samples via ATR FT-IR and dynamic mechanical analysis. Designed and built a custom water-filled impedance tube for underwater impedance testing.

Postgraduate Research Assistant

Department of Polymer Science and Rubber Technology, CUSAT · Ernakulam, Kerala, India · Dec 2020 - Aug 2022

Worked on moldable osteoinductive bone graft formulations from bioglass-intercalated quaternised chitosan–sodium alginate polyelectrolyte complexes. Synthesised and characterised PECs, performed in vitro biological characterisation (MTT, Actin–DAPI) and bone regeneration marker studies. Also worked on phosphorylated nanocellulose and intrinsically radiopaque bone cements.

Project Trainee

Department of Biological Sciences and Bioengineering, IIT Kanpur · Kanpur, Uttar Pradesh, India · Dec 2018 - May 2019

Developed photo-polymerising injectable hydrogels for improved cartilage repair. Methacrylated gelatin, carboxymethyl cellulose and silk fibroin to give Gel-MA, CMC-MA and Silk-MA. Photo-crosslinked in the presence of LAP under blue light. Characterised for cell biocompatibility, degradation, compressive strength and porosity.

PUBLICATIONS

1. **Early Biomimetic Radiopaque Phosphorylated Nanocellulose Intercalated Bone Grafts for Setting Composites for Periodontal Regeneration.**

TECHNICAL SKILLS

Material synthesis

Polymer synthesis & modification · Biopolymers and biopolymers from agro-waste · Nanocellulose extraction and phosphorylation · Bioglass, hydroxyapatite, brushite · Self-setting bone cements · Quantum dots and radiopaque fillers · Interpenetrating polymer networks · Photo-crosslinkable hydrogels (Gel-MA, CMC-MA, Silk-MA)

Characterisation

ATR FT-IR · Thermogravimetric analysis · Dynamic mechanical analysis · FE-SEM · Universal testing machine · In vitro biomimetic mineralisation · Swelling and degradation studies · UV/Vis spectrophotometry

Biology & computation

L929 fibroblast cell culture · MTT cytotoxicity assay · Actin–DAPI staining · Bone regeneration marker studies · Underwater impedance testing · Python, OriginPro, image analysis

Languages

English (professional) · Malayalam (native) · Hindi (communicative)